Fiscal Year:	FY 2022	**Task Last Updated:**	FY 04/18/2022
PI Name:	Goree, John Ph.D.		
Project Title:	Nonlinear Wave Experiments in Dusty Plasmas		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Complex fluids		
Joint Agency Name:	National Science Foundation	**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	john-goree@uiowa.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	319-335-1843
Organization Name:	University of Iowa		
PI Address 1:	Department of Physics and Astronomy		
PI Address 2:	210 Van Allen Hall		
PI Web Page:			
City:	Iowa City	**State:**	IA
Zip Code:	52242-1403	**Congressional District:**	2
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2016 Physical Sciences NNH16ZTT002N: Fundamental Physics – NASA/NSF Partnership on Science of Dusty Plasmas: Utilizing the PK-4 Facility on board the ISS
Start Date:	06/14/2017	**End Date:**	12/31/2023
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
Contact Monitor:	Callas, John	**Contact Phone:**	
Contact Email:	john.l.callas@jpl.nasa.gov		
Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date changed to 12/31/2023 per J. Callas/JPL (Ed., 11/14/22) NOTE: End date changed to 12/13/2021 per U. Israelsson/JPL (Ed., 5/25/21) NOTE: Change in start and end dates to 6/14/2017-6/13/2021 (from 4/15/2017-4/15/2021) per I. Hahn/JPL and PI (Ed., 1/19/18)		
Key Personnel Changes/Previous PI:	During the reporting period, Co-I Bin Liu left the University of Iowa, and is no longer a Co-I.		
COI Name (Institution):			
Grant/Contract No.:	JPL 1573629		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	A flight experiment will be performed and analyzed in a study of nonlinear synchronization of waves. Synchronization is a cooperative phenomenon for oscillators or waves. A signature of synchronization is a frequency that shifts when an external modulation is applied. Compared to point-like oscillators, waves are more complicated due to their spatio-temporal behavior and dispersion properties. An understanding of synchronization is lacking for waves of all kinds, including those in dusty plasmas. The experiment will use dusty plasmas, also called "complex plasmas." These are mixtures of small particles of solid matter, electrons, ions, and neutral gas. The small solid particles obtain large electric charges, so that they exhibit collective behavior. They are electrically confined, so that they do not contact any boundaries. Microgravity enables this experiment by suppressing sedimentation, thereby allowing the waves to propagate distances of many wavelengths, with relatively uniform background conditions. This experiment is based on previous ground-based experiments, where waves propagated only a few wavelengths. Our flight experiment will be performed using “Plasma Kristall 4” (PK-4), which is a facility in the Columbus Module of the International Space Station (ISS), built by the European Space Agency (ESA). SCIENTIFIC/TECHNICAL MERITS The main scientific topic is the synchronization of a dust acoustic wave (DAW). A DAW is a low-frequency compressional wave, which involves charge-separation from electrons and ions. The DAW is self-excited by an ion flow, and it grows naturally to a large amplitude so that nonlinear effects occur readily. Synchronization experiments are performed by applying electrical waveforms to an electrode to modulate the dusty plasma sinusoidally. The diagnostic for detecting the wave motion is video imaging using particle observation cameras. Image data will be analyzed to prepare space-time diagrams, which provide a spatial and temporal description of the wave's amplitude as it propagates. These space-time diagrams are the foundation for further analysis methods. In support of our experiment, a theoretical effort will be undertaken at Iowa. The goal is to contribute to the development of a theory for synchronization of propagating waves, to compare to the experiment. The theoretical effort at Iowa will focus on numerical simulations using a hybrid model. Molecular dynamics simulation will also be performed. The main theoretical effort, which will center on analytic theory, will be done by theorist elsewhere, and we will collaborate with them. The feasibility of our flight experiment is based on our previous ground-based experiments, which confirm that synchronization of the DAW occurs. These ground-based experiments also revealed unexplained phenomena, including a nonharmonic synchronization state, which we will investigate with our flight data. In the unlikely event that our main experiment with waves does not yield useful results, we have a plan for a backup experiment to investigate mobility-limited motion of particles having a different size. BROADER IMPACTS This research has an overlap with nonlinear dynamics as well as other plasma physics areas outside of dusty plasmas. Research results will be disseminated in journals, conferences, a website, and Twitter. A K-12 outreach presentation will be made at an elementary school.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	N/A
Task Progress:	During the six-month-extension period, we prepared and published a new paper for publication in a peer-reviewed journal. This paper reports our new results from our flight experiment performed in "Plasma Kristall 4” (PK-4) on the International Space Station (ISS). A highlight from this paper is an observation of nonlinear synchronization in a 3D dusty plasma on the ISS. We found dust motion exhibited two distinctive frequency clusters that oscillated at harmonic frequencies. This finding marks a milestone in achieving our scientific goals for this project.
Bibliography Type:	Description: (Last Updated: 01/23/2025)
Articles in Peer-reviewed Journals	Liu B, Goree J, Pustylnik MY, Thomas HM, Fortov VE, Lipaev AM, Usachev AD, Petrov OF, Zobnin AV, Thoma MH. "Time-dependent shear motion in a strongly-coupled dusty plasma in PK-4 on the International Space Station (ISS)." IEEE Transactions on Plasma Science. 2021 Sep;49(9):2972-78. https://doi.org/10.1109/TPS.2021.3100300 , Sep-2021
Articles in Peer-reviewed Journals	Liu B, Goree J, Schütt S, Melzer A, Pustylnik MY, Thomas HM, Fortov VE, Lipaev AM, Usachev AD, Petrov OF, Zobnin AV, Thoma MH. " Nonlinear wave synchronization in a dusty plasma under microgravity on the International Space Station (ISS)." IEEE Transactions on Plasma Science, 2021 Dec;49(12):3958-62. https://doi.org/10.1109/TPS.2021.3123556 , Dec-2021